

Rocks and minerals

The chemistry of Earth is dominated by the huge variety of rocks and minerals that shape the landscape around us.

There are thousands of different kinds of rocks and minerals. What they are like depends on the chemical elements they contain, and the way these elements are grouped together. A rock is a mixture of different minerals, arranged as billions of tiny grains. Each mineral is usually a compound of two or more elements chemically bonded together. Many of these form beautiful crystals. Sometimes, a mineral is an element in its raw form—such as copper or gold.

Most of the ocean floor is made of **igneous basalt rock**, much younger than most rocks on land.

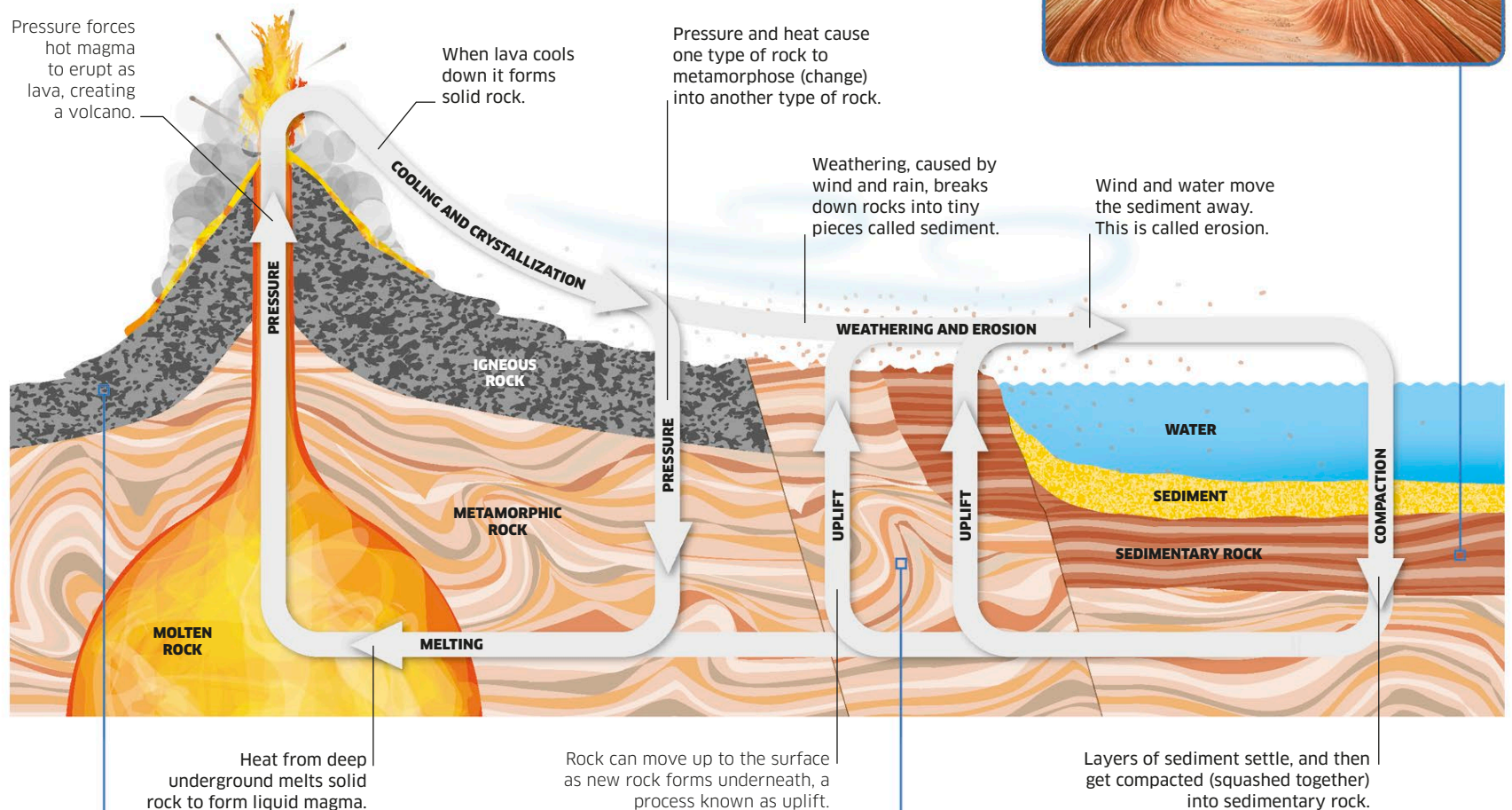
Sedimentary rock

Fragments of rock broken away by weathering and erosion join together to form sedimentary rocks, such as sandstone (below) and limestone. The fragments gather in layers at the bottom of lakes and oceans, and get compacted and cemented together under their own weight. Eventually, uplift pushes this rock up to the surface.



The rock cycle

Solid rocks look like they must stay the same forever, but in fact they change over thousands or millions of years. Some melt under the influence of Earth's internal heat and pressure. Others get eroded by wind and rain. The three main forms of rock are linked in a cycle that changes one form into another. The cycle is driven slowly, but inevitably, by a set of dramatic movements deep within the Earth.



Igneous rock

The interior of the Earth is so hot it melts solid rock, forming a liquid called magma. When magma cools down it solidifies and crystallizes to form igneous rock, such as granite (formed underground) and basalt (seen left) from lava erupted from volcanoes.



Metamorphic rock

Rocks that get buried deep underground are squeezed and heated under pressure. But instead of melting the rock, this rearranges its crystals to form metamorphic rock. For example, buried limestone changes into marble, as in this cave.